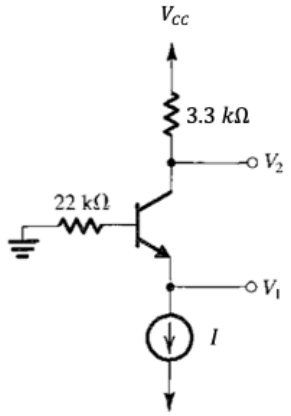
1) For the circuit below, find the indicated voltages for each value of the current I and record in the table. Use $\beta = 100$. Show all steps.



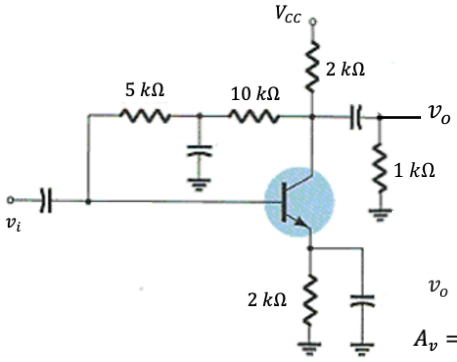
I=1 mA: active mode.

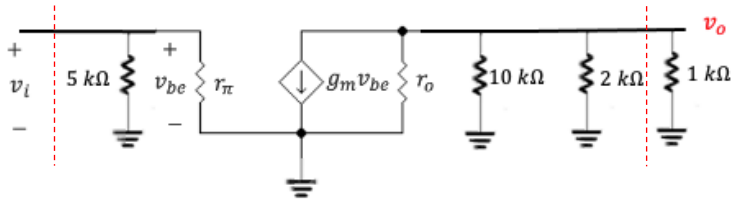
$$I_E = 1\,mA,\ I_C = \frac{100}{101}1\,mA,\ I_B = \frac{1}{101}1\,mA \rightarrow V_B = 0 - 22k(I_B),\ V_1 = V_B - 0.7,\ V_2 = V_{CC} - 3.3(I_C)$$

I=4 mA: saturation mode.

$$I_E = 4\,mA = I_C + I_B = \frac{V_{CC} - V_2}{3.3} + \frac{0 - (V_2 + 0.5)}{22} \rightarrow \text{solve for } V_2,\ V_1 = V_2 - 0.2$$

2) Analyze the amplifier circuit below with $\beta = 100$ and V_A=50 V. Find the amplifier voltage gain, current gain and input and output resistances. Use the (pi) model for the BJT and <u>do not</u> ignore r_o. Show all steps and calculations. Enter results in table.





$$v_i = v_{be}$$

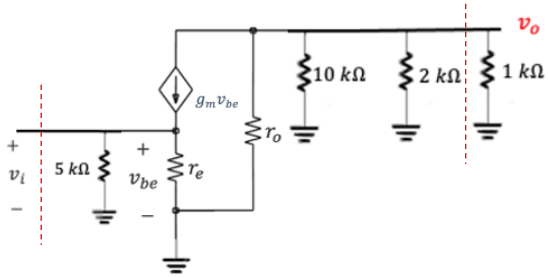
$$v_o = -g_m v_{be}(r_o \,//\, 10k \,//\, 2k \,//\, 1k)$$

$$A_v = \frac{v_o}{v_i} = -g_m\,(r_o \,//\, 10k \,//\, 2k \,//\, 1k)$$

$$R_i = (r_\pi \,//\, 5k)$$

$$R_o = (r_o \,//\, 10k \,//\, 2k)$$

$$A_i = A_v \frac{R_i}{1k}$$



3) Implement the logical function F using minimum CMOS logic. Perform transistor sizing such that the performance of the circuit matches the basic CMOS inverter with transistor size $n = 1$ and $p = 3$. Find the total circuit area for L=0.1 μm. Assume that the inputs and their complements are available.

$$F = \overline{AB} + CD = \overline{AB(\bar{C} + \bar{D})} = \overline{AB\bar{C} + AB\bar{D}}$$

$$F = \overline{AB\bar{D}} + C = \overline{AB\bar{C}D}$$

$$F = \overline{AD} + \overline{BC} = \overline{ABCD}$$

$$F = \bar{A} + BCD = \overline{A(\bar{B} + \bar{C} + \bar{D})}$$

$$F = \bar{B} + \overline{AD} + C = \overline{AB\bar{C}D}$$